Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2025-2026(EVEN)

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

SHEIKH WASIMUDDIN

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Shiky
6/3/26

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Session	**2024-25 (EVEN)**	**Course Name**	**Mobile and Web Application for IOT Lab**
Semester	**6**	**Course Code**	**23IOT1624**
Roll No	**63**	**Name of Student**	**SHEIKH WASIMUDDIN**

Practical Number	**08**
Course Outcome	1. Understand and apply basic Kotlin programming constructs including variables, control structures, functions, null safety, and object-oriented concepts for Android app development. 2. Design and develop responsive user interfaces using XML layout files, ViewGroups, and widgets in Android Studio, adhering to Material Design principles. 3. Implement event handling and application logic using Kotlin to capture user input, manage app state, navigate between screens, and interact with Android system components like Toasts and Dialogs 4. Integrate learned concepts to develop a mini Android application that demonstrates use of ViewModel, multi-screen navigation, and proper UI/UX design for real-world problem-solving
Aim	Manage UI state using ViewModel.
Problem Definition	Develop an Android application that maintains a counter value using ViewModel and LiveData. The counter should increase when a button is clicked, and the value should remain unchanged even after configuration changes such as screen rotation
Theory (100 words)	Managing UI data properly is essential in Android applications to ensure that the user experience remains smooth during configuration changes such as screen rotation or device orientation changes. When an activity is recreated, the UI components may lose their data if it is not handled correctly. To overcome this issue, Android provides Architecture Components like ViewModel and LiveData. The ViewModel class is responsible for storing and managing UI-related data so that it survives configuration changes. LiveData is an observable data holder that allows the UI to automatically update whenever the data changes. Together, ViewModel and LiveData help maintain application state, improve performance, and simplify the process of updating the user interface efficiently.

Procedure and Execution (100 Words)	**Step for Implementation:** 1. Open Android Studio and create a new Android project. 2. Design the layout with TextView and Button in activity_main.xml. 3. Create a ViewModel class to store the counter value. 4. Use MutableLiveData in the ViewModel to hold the counter data. 5. Implement a method in ViewModel to increase the counter value. 6. Initialize the ViewModel inside MainActivity. 7. Observe the LiveData object to update the TextView automatically. 8. Set a click listener on the button to call the increment function. 9. Run the application on an emulator or device. 10. Rotate the screen to verify that the counter value remains unchanged.

Code:

1. Manifest

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:label="practical8_iot63"
        android:supportsRtl="true"
        android:theme="@style/Theme.Material3.DayNight.NoActionBar">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>

                <category
android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>

        </activity>
```

```
    </application>

</manifest>
```

2. MainActivity.kt

```
package com.example.practical8_iot63

import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.activity.viewModels
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    private val viewModel: CounterViewModel by viewModels()

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val textCounter = findViewById<TextView>(R.id.textCounter)
        val btnIncrease = findViewById<Button>(R.id.btnIncrease)

        viewModel.counter.observe(this) {
            textCounter.text = it.toString()
        }

        btnIncrease.setOnClickListener {
            viewModel.increaseCounter()
        }
    }
}
```

3. ActivityMain.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
```

```
    <TextView
        android:id="@+id/textCounter"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="0"
        android:textSize="30sp"/>

    <Button
        android:id="@+id/btnIncrease"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Increase Counter"/>

</LinearLayout>
```

4. SecondActivity.kt

```
package com.example.practical8_iot63

import androidx.lifecycle.LiveData
import androidx.lifecycle.MutableLiveData
import androidx.lifecycle.ViewModel

class CounterViewModel : ViewModel() {

    private val _counter = MutableLiveData<Int>(0)
    val counter: LiveData<Int> = _counter

    fun increaseCounter() {
        _counter.value = (_counter.value ?: 0) + 1
    }
}
```

Output:

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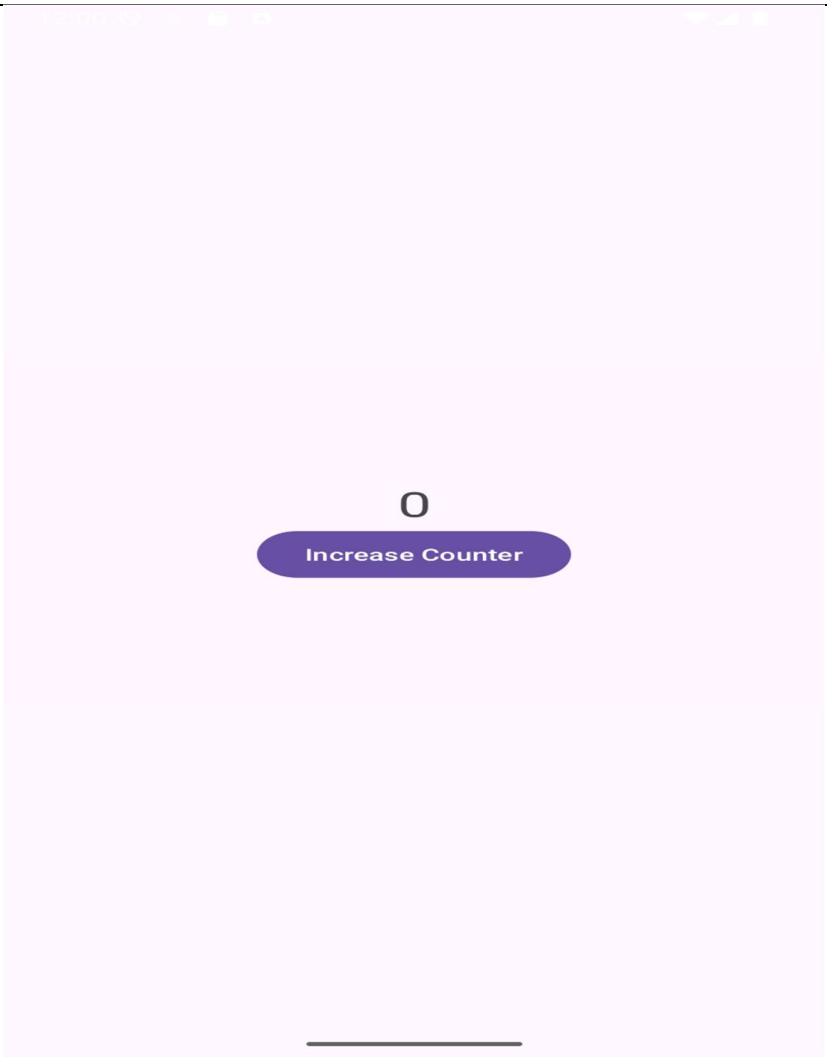

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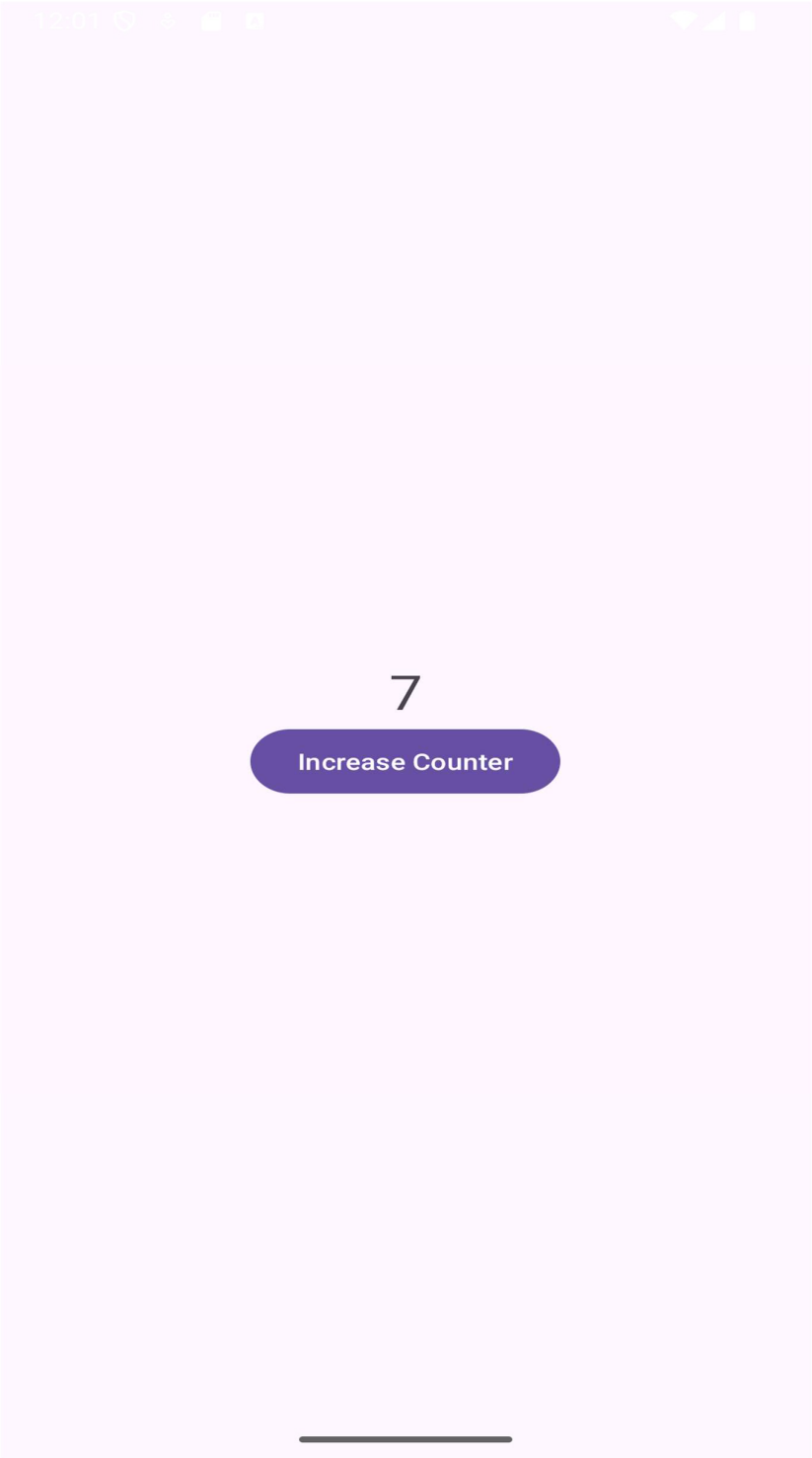
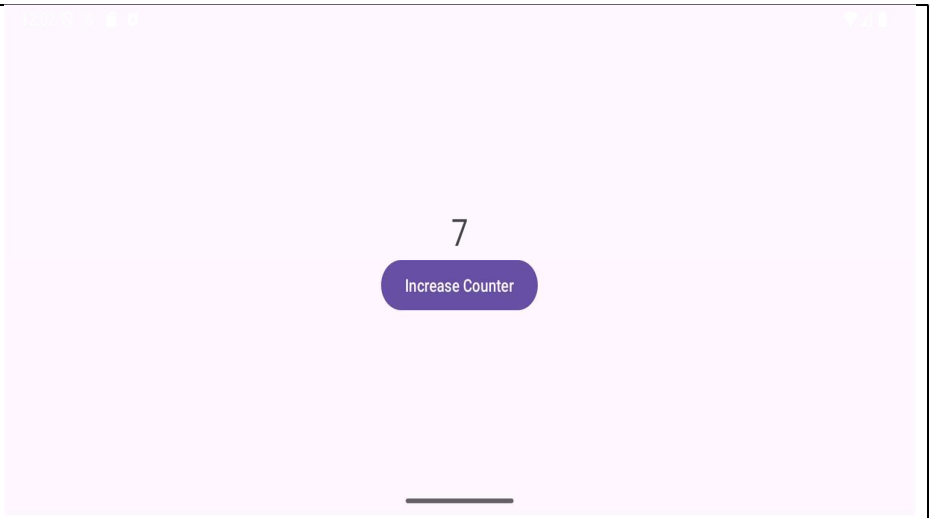
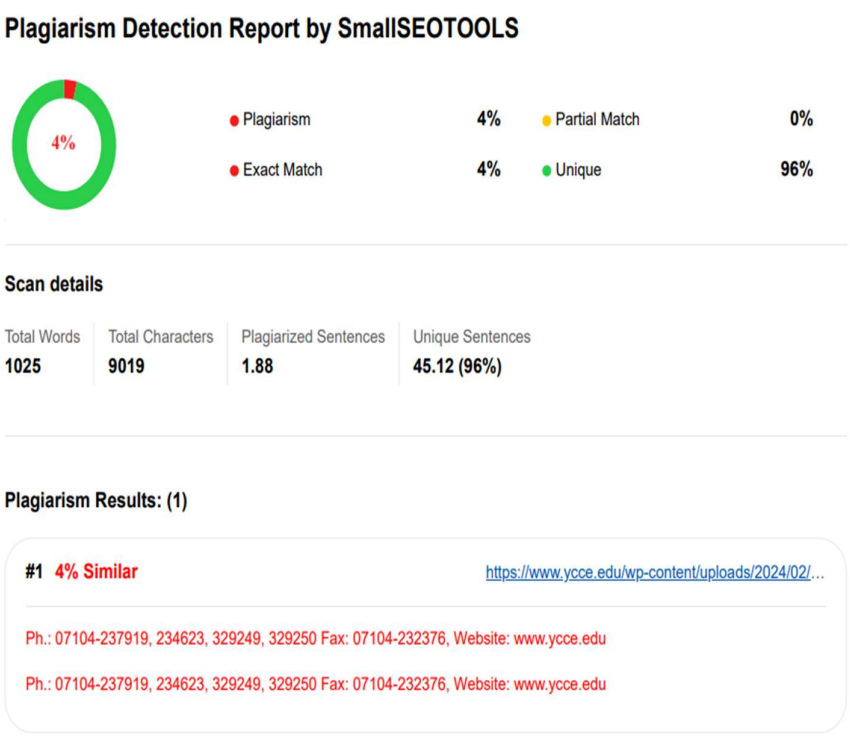


	
Output Analysis	The developed Android application successfully demonstrated proper UI state management using ViewModel and LiveData. When the user interacted with the button, the counter value increased and the updated value was instantly reflected on the screen. The LiveData component observed the data changes and automatically updated the user interface without requiring manual refresh. To verify the functionality, the device orientation was changed during execution. Even after the activity was recreated, the counter value remained the same, indicating that the ViewModel retained the data successfully. This experiment confirmed that Android Architecture Components effectively handle configuration changes, maintain UI data, and enhance the stability and performance of the application.
Link of student Github profile where lab assignment has been uploaded	https://github.com/sheikhwasimuddin
Conclusion	This experiment illustrated the importance of handling UI state in Android applications using modern architecture components. By implementing ViewModel, the application was able to retain important data even when the activity was recreated due to configuration changes like screen rotation. LiveData played a key role in observing data changes and updating the user interface automatically without additional coding effort. This approach reduced the chances of data loss and made the application more

	reliable and responsive. Overall, the experiment provided a clear understanding of lifecycle-aware components and demonstrated how they improve application stability, maintain UI consistency, and support efficient Android application development.
Plag Report (Similarity index < 12%)	 **Plagiarism Detection Report by SmallSEOTOOLS** 4% Plagiarism 4% Partial Match 0% Exact Match 4% Unique 96% **Scan details** Total Words 1025 Total Characters 9019 Plagiarized Sentences 1.88 Unique Sentences 45.12 (96%) **Plagiarism Results: (1)** #1 4% Similar https://www.ycce.edu/wp-content/uploads/2024/02/... Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Date	**06/03/2026**